

Diwali Sales Analysis Using Python :

IILM University School of Computer Science & Engineering

Student Name: Harsh Raj

Enrollment No: CS-2341780

Section: 4CSE12

Program & Semester: B.Tech CSE(DS&BD), 7th Sem

Organization: Upskill Campus



Introduction & Problem Statement



Introduction

Diwali is one of India's biggest festive shopping seasons, driving a significant surge in consumer spending. Purchase patterns vary by **gender, age, occupation, location, and product category**. Data analysis helps businesses understand these patterns and make informed decisions.

Problem Statement

Without data-driven insights, businesses miss opportunities for **targeted marketing, inventory planning, and sales optimisation**. This project addresses that gap using Python-based EDA.

Objectives

1

Clean & Preprocess

Handle missing values, remove unnecessary columns, and prepare the raw Diwali sales dataset for analysis.

2

Exploratory Data Analysis

Perform EDA across gender, age group, state, occupation, and product category using Python.

3

Identify Key Segments

Find the highest-contributing customer segments, top-performing regions, and best-selling product categories.

4

Generate Insights

Produce actionable recommendations to support marketing strategy and business growth.

Dataset & Technologies

Dataset

Diwali Sales Dataset in CSV format, containing customer demographics and transaction-level sales records.

Technologies Used

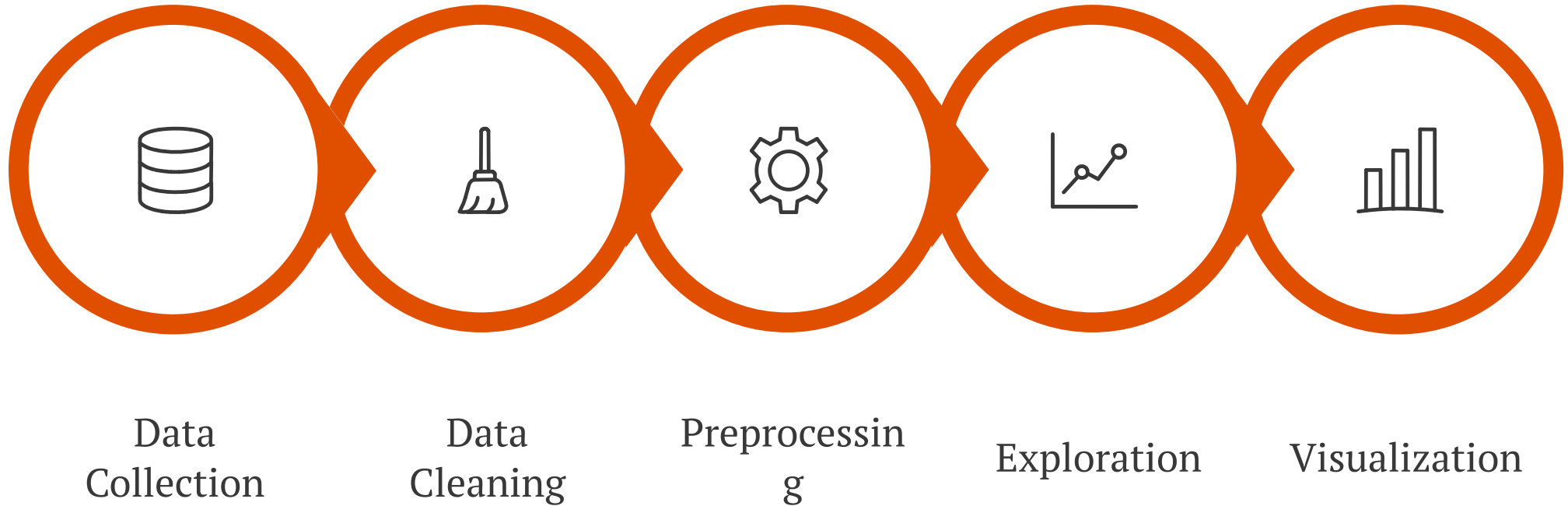


Python	Core data analysis
Pandas	Data cleaning & manipulation
NumPy	Numerical operations
Matplotlib	Basic visualisation
Seaborn	Statistical graphs
Jupyter Notebook	Project environment

Data Preparation

Missing values removed · Unnecessary columns dropped · Data types adjusted · Index reset

Methodology



The analysis covers five key dimensions: **gender-wise spending**, **age-group spending**, **state-wise sales**, **occupation-wise sales**, and **product-category performance** — each explored through visualisation and statistical summaries.

Exploratory Data Analysis

Five dimensions of the Diwali sales dataset were analysed using Python visualisation libraries.



Gender

Spending comparison between male and female customers



Age Group

Highest-spending age bracket identification



State

Sales distribution across Indian states



Occupation

Purchasing power across professions



Product Category

Most popular product categories by sales



Insert your 5 EDA graph screenshots here for the final presentation.

Key Results & Findings



Gender

Women contributed significantly more to total sales than men.



Age

Customers aged **26–35** recorded the highest spending across all categories.



Geography

Uttar Pradesh, Maharashtra, and Karnataka were the top-performing states.



Occupation

Customers from **IT, Healthcare, and Aviation** showed the strongest purchasing power.



Products

Clothing & Apparel, Electronics & Gadgets, and Home Decor & Appliances led in sales volume.

Business Insights

How businesses can apply these findings:



Customer Targeting

Focus festive campaigns on **women** and the **26–35 age group** — the highest-spending segments.



Regional Strategy

Increase promotions in **Uttar Pradesh, Maharashtra, and Karnataka** and other high-performing urban markets.



Product Strategy

Maintain strong inventory for **Clothing, Electronics, and Home Decor** during the festive season.



Marketing

Design targeted festive discounts and personalised offers for high-value customer segments.



Inventory Planning

Use historical purchasing patterns to align stock levels with anticipated customer demand.

Conclusion & Future Scope

Conclusion

The project successfully analysed Diwali customer purchasing behaviour using Python. EDA revealed meaningful demographic, regional, and product-level patterns. These findings can directly support **marketing, inventory, and customer-targeting decisions** for festive-season businesses.

Future Scope

- Interactive dashboards using **Power BI, Tableau, or Streamlit**
- Customer segmentation and clustering
- Sales forecasting models
- Market Basket Analysis
- RFM Analysis for customer value scoring
- Real-time e-commerce data integration

Thank You

Presented By: Harsh Raj